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# Easily construct, convert or build Likert items from vectors or data frames

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## Executive Summary

This proposal outlines the development of an open-source R package for converting or constructing Likert items from vectors or data frames, particularly from non-standard data structures. The proposed package eliminates fragmented, manual workarounds for non-standard data structures by introducing a reproducible workflow that mitigates data manipulation risks. Operating on a development model of 10 hours per week, the project features a 15-week timeline and a budget of USD 8,500. The timeline is divided into a three-week inception phase; a ten-week coding and testing phase, and initial CRAN submission with concurrent sensitization workshop; and a final two-week phase for incorporating community feedback and project closure. The budget funds 150 hours of core development labor at USD 50 per hour, documentation hosting, and a workshop for faculty and students at the Federal University of Technology, Minna, Nigeria. Upon completion, the package and its continuous updates will be maintained publicly in GitHub and CRAN repositories.

## Signatories

### Project team

Professor Job Nmadu is leading the task and would deliver on the targets.

### Contributors

Dr. Halima Sallawu of Federal University of Technology, Minna, Nigeria and Professor Job Nmadu have participated in data collections necessitating this project.

## Consulted

## The Problem

The CRAN packages for Likert-type data workflow, viz, Mollazehi (2025); Winzar (2026) and Bryer and Speerschneider (2025), did not encompass seamless *type coercion* to alternative structures for robust downstream processing, but, facilitates standard Likert items workflows. They work with conventional, clean ordinal data formats and lack the native architectural framework required for structural transformations.

Empirical workflow frequently encounter non-standard data which require programmatic modification; such as mapping items/scales into ordinal factors, turning multi-point data into binary indicators, or synthesizing Likert scales from continuous data. The existing workflow cannot execute these natively, forcing researchers to rely on fragmented, manual workarounds and ad-hoc scripting. This fragmentation disrupts analysis, introduces data entry risks, and leads to loss of man-hours. This project proposes an open-source R package for automated type coercion and dynamic vector mapping to systematically bridge this gap while preserving data integrity, and handling non-standard data natively.

## The proposal

Provide robust algorithms and optimized workflows for the following concepts:

- Binary Coercion: Convert multi-point Likert scores to binary variables
- Ordinal Factor Mapping: Convert numeric or text data into clean, ordered or Likert scores
- Text-to-Score Mapping: Parse and map Likert text responses into numeric score sequences
- Vector distribution/description: Generate summary, distribution or structural profiles of Likert vectors
- Index Weight Construction: convert multi-item Likert data to composite indicators or indices

## Overview

The proposal is to create an R package optimized for Likert-type data transformations.

## Detail

### Minimum Viable Product

The package, focusing on Likert-type data structures for construction and conversion, is hosted on GitHub.

### Architecture

To minimize external dependencies, Base R functions will be used. The core algorithm revolves around *vector coercion*, type casting, and matrix mapping but where feasible **tidyverse** higher-level functions will be called.

## Assumptions

While the project's is technically sound, proposing a 15-week timeline introduces some risks identified below with their mitigation:

- Unanticipated disruptions of daily time budget can be mitigated by using weekends
- `devtools::check()` across multiple OS can limit rejection or delays from CRAN checks or reviewers
- Minimal external dependencies and unit tests can mitigate breaking codes from internal conversion logic
- Version control and testing offline can mitigated power outages or unstable internet
- conducting workshops virtually can mitigate natural disasters (floods/earthquakes/hurricanes) and global crises (war/pandemics)

## External dependencies

The proposed package will rely on Base R and `tidyverse` established routines and workflows.

## Project plan

### Start-up phase

The open-source R package, under GPL licence, is aimed at establishing the GitHub repository and integrating it with the local machine workflows. The first priority is designing the system's architecture and initial abstract classes.

### Technical delivery

Procedure for continuous integration, stakeholder feedback and updates by R Core teams are in place after package deployment to GitHub and CRAN repositories.

### Other aspects

Announcement posts and delivery blogs for continuous feedback while social media posts will be used for stakeholder engagement.

## Budget & funding plan

| Milestone              | Duration<br>(weeks)[1] | Deliverable/Tangible Products                                                                                            | Budget<br>(USD) |
|------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------|
| Inception and take-off | 3                      | Establish GitHub repository and local machine integration. Defining the system architecture and initial abstract classes | 1500            |

| Milestone                 | Duration<br>(weeks)[1]                      | Deliverable/Tangible Products                                                                                                                                                  | Budget<br>(USD) |
|---------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Coding and testing        | 10                                          | Set up the functions and algorithms, build the website and GitHub pages. Pass integration tests with local backends and CRAN submission                                        | 5000            |
| Sensitization Workshop    | concurrently with <i>Testing and Coding</i> | Conduct a sensitization workshop and stakeholder engagement for faculty and students at the Federal University of Technology, Minna. Drive publicity over social media handles | 1000            |
| Final release and closure | 2                                           | Incorporate stakeholder feedback, resolve CRAN review comments, clean up repository branches, and achieve release on CRAN                                                      | 1000            |
| Total                     | 15                                          |                                                                                                                                                                                | 8500            |

[1] 10 hours are devoted per week

## Success

### Definition of done

Functional, CRAN-accepted R package with comprehensive documentation and community onboarding

### Measuring success

Following automated testing, stakeholder feedback, and R Core Team updates, the code are deployed to GitHub and CRAN.

## References

- Bryer, Jason, and Kimberly Speerschneider. 2025. *Likert: Analysis and Visualization Likert Items*. <https://doi.org/10.32614/CRAN.package.likert>.
- Mollazehi, Mohammad. 2025. *LikertEZ: Easy Analysis and Visualization of Likert Scale Data*. <https://doi.org/10.32614/CRAN.package.LikertEZ>.
- Winzar, Hume. 2026. *LikertMakeR: Synthesise and Correlate Likert Scale and Rating-Scale Data Based on Summary Statistics*. <https://doi.org/10.32614/CRAN.package.LikertMakeR>.